Anh (Andre) Tran

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Education

University of Cincinnati Expected May 2028
Bachelor of Science in Computer Engineering, Minor in Computer Science *GPA: 3.77/4.00*
Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design & Development, Cybersecurity, Digital Design & Digital Systems, Artificial Intelligence, Context Engineering

Experience

University of Cincinnati Department of Computer Science 08/2026 - Present
AI Researcher *Cincinnati, OH*
- Cut voice-resynthesis time 15x (5.5s to 0.36s) via **LPC** pole-warping and **LP-PSOLA**, profiling pitch tracking to 89% of runtime, then swapping in a 4x decimated **autocorrelation** search over an 8x option that halved accuracy.
- Fixed an **LPC** bug destroying 97.7% of output frames with a **Schur-Cohn** test, 100x cheaper than root-finding.
- Found and fixed 3 measurement bugs that had silently corrupted previously collected study data.

Commma - Pace your Code (commma.dev) 05/2026 - Present
Founder & Lead Engineer *Cincinnati, OH*
- Building **commma.dev**, a coding-activity tracker with 700+ downloads on **VS Code** and **Open VSX**, as a **TypeScript** monorepo (extension, **Hono REST** API, **PostgreSQL**, **React** SPA) on **AWS** via **Terraform**.
- Made coding-time totals hold under concurrency and crashes: 2,390 sessions vs 1,220 expected and 35/40 SIGKILL trials losing **Redis** leaderboard time, both to 0 with a **PostgreSQL** advisory lock and a **transactional outbox**.
- Verified **privacy** (no editor content sent) with **fast-check** property tests: 9,000 randomized runs over 361,455 synthetic edit events found 0 leaks, and all 4 deliberately injected leaks were caught.
- Cut ingest's privacy lookup p50 70% (42ms to 13ms) with a **Redis** cache, then closed a 60s stale-cache race.

Caphne - Study Buddies Matchmaking (caphne.co) 01/2026 - 08/2026
Lead Software Engineer *Remote*
- Cut API response time from 60s to 10s by engineering a distributed, real-time chat system with **Socket.IO** scoped to per-match rooms, sustaining 1.7K+ requests/minute across 30 days of production traffic.
- Led a cross-functional team of 6 engineers through sprint planning, code reviews, and production deployments of a full-stack **Nuxt 3**, **Vue**, and **Node.js** platform, scaling to 400+ active users in production.
- Architected **AWS** infrastructure as code (**EC2**, **CloudFront**, **S3**, **Route 53**) via **Terraform** with **OIDC**-federated credentials, then root-caused and fixed 2 production incidents same-day from raw error output.
- Deleted two speculative database tables from match rate-limiting, raising daily match supply 7x (3/3hr to 20/day).

Projects

Kiln-IDP (Live) | *Go, Java (Spring Boot), Kubernetes, ArgoCD, Istio, Kafka, PostgreSQL, OpenTelemetry*
- Engineered a cost-aware **Kubernetes** scheduler plugin that placed zero latency-class workloads on spot capacity, validated across 57,012 replayed placements on a 12,477-machine public cluster trace.
- Designed a canary controller using a Wald sequential probability ratio test and **CUSUM** change detection, catching 100% of 1,000 seeded regressions at a 2.8% false-rollback rate on a live **Istio** service mesh.
- Replaced ticket-and-**Terraform** provisioning with a self-service path reaching Ready at 11s p50 across 6 subsystems.
- Lifted a **controller-runtime** 20qps/30burst limit, cutting 1,000-rollout promotion from a 12-min timeout to 3m58s.

Draft-Thinker (Live) | *Go, OpenAI API, Qdrant, Redis, Prometheus, Grafana, Docker*
- Cut inference cost 82.9% (bootstrapped 95% CI: 79.6-86.0%) while holding accuracy at 98.1% by designing a **machine learning** entropy router in **Go** using top-k logprobs from the **OpenAI API** at inference time.
- Diagnosed a confidence-threshold baseline's 61.5% cost-reduction ceiling to an **OpenAI** logprob-reporting artifact affecting 65.5% of creative prompts; entropy routing structurally avoids this failure mode.
- Found cancelled speculative calls would cost 80-88% of the all-heavyweight bill, recommending speculation stay off.

Ferrox (Live) | *Rust, C++, Systems Programming, Atomics, Mmap, Criterion*
- Built a low-latency, price-time matching engine in **Rust** with fixed-point prices, trading a 9-22% insert regression for a 67-85% cut in cancel latency via a **HashMap**-to-**BTreeMap** swap, sustaining 4.7M simulated orders/sec.
- Ported the lock-free ring buffer to **C++20**, confirming **Acquire/Release** elides all x86 fences versus a 5.4-6.0x SeqCst penalty, then isolated **Rust**'s 1.5-3.5x single-threaded gap to its runtime bounds-check overhead.
- Deployed a containerized service to **AWS Fargate**, measuring the NLB at 70% of total hourly cost.
- Fixed two silent-corruption bugs via **crash-consistency fault injection**, zeroing 714 failures across 92,525 records.

Technical Skills

Languages: Go, JavaScript/TypeScript, SQL, Python, Rust, C/C++, C#, Java, HTML/CSS
Technologies: Nuxt.js, Vue.js, Next.js, React, Node.js, Express.js, Spring Boot, Hono, Kubernetes, Kafka, Git, Redis, LangChain/LangGraph, AWS (Amazon Web Services), Docker, PostgreSQL, MongoDB, MySQL, FastAPI, Django, ChromaDB, scikit-learn, NumPy, Socket.IO, GitHub, Grafana, Prometheus, Terraform, CI/CD, Agile/Scrum, Linux